Chart Chooser

Start with the question. Find the chart. Copy one line.

DATAMOUS
Cheat Sheet
Every line tested on matplotlib, seaborn and plotly

FOUR STEPS 1 Write the message in one sentence > 2 Find your question > 3 Draw it > 4 Run the checklist (page 2)

BEFORE YOU START

Imports
```
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
```

New chart
```
fig, ax = plt.subplots()
```

Per product
```
t = (df.groupby('product', as_index=False)
    ['revenue'].sum()
    .sort_values('revenue', ascending=False))
```

Per month
```
m = (df.groupby('month', as_index=False)
    ['revenue'].sum())
```

Two groups
```
g = df.groupby(['region', 'segment'],
    as_index=False)['revenue'].sum()
```

Same, wide
```
w = g.pivot(index='region',
    columns='segment', values='revenue')
```

`df` has one row per sale: `month`, `product`, `region`, `segment`, `price`, `units`, `revenue`.

Which is biggest? SORTED BARS

Rank items by size, biggest first.

matplotlib
```
ax.barh(t['product'], t['revenue'])
ax.invert_yaxis()
```

seaborn
```
sns.barplot(data=t, x='revenue', y='product')
```

plotly
```
px.bar(t, x='product', y='revenue')
```

WATCH Bars start at zero. `invert_yaxis()` puts the first row on top.

How many of each? COUNT BARS

Count the rows in each category.

matplotlib
```
c = df['region'].value_counts()
ax.bar(c.index, c)
```

seaborn
```
sns.countplot(data=df, x='region')
```

plotly
```
px.histogram(df, x='region')
```

WATCH `countplot()` counts rows. Have counts already? `barplot()`.

Two groupings? GROUPED BARS

Compare one number across two categories.

matplotlib
```
w.plot.bar(rot=0, ax=ax)
```

seaborn
```
sns.barplot(data=g, x='region', y='revenue',
    hue='segment')
```

plotly
```
px.bar(g, x='region', y='revenue',
    color='segment', barmode='group')
```

WATCH Four colors at most. More groups? Gray plus one highlight.

Spread by group? BOX PLOT

Compare the spread of values between groups.

matplotlib
```
df.plot.box(column='revenue', by='region',
    ax=ax)
```

seaborn
```
sns.boxplot(data=df, x='region', y='revenue')
```

plotly
```
px.box(df, x='region', y='revenue')
```

WATCH The line is the median. A box hides how many values it holds.

Change over time? LINE

Show how a number rises and falls.

matplotlib
```
ax.plot(m['month'], m['revenue'])
```

seaborn
```
sns.lineplot(data=m, x='month', y='revenue')
```

plotly
```
px.line(m, x='month', y='revenue')
```

WATCH Axis above zero? State the size of the change in the title.

How spread out? HISTOGRAM

See where most values fall, and the odd ones.

matplotlib
```
ax.hist(df['revenue'], bins=20)
```

seaborn
```
sns.histplot(data=df, x='revenue', bins=20)
```

plotly
```
px.histogram(df, x='revenue', nbins=20)
```

WATCH Try a few bin counts. The shape can change with them.

Move together? SCATTER PLOT

Check if two numbers rise and fall together.

matplotlib
```
ax.scatter(df['price'], df['units'])
```

seaborn
```
sns.scatterplot(data=df, x='price', y='units')
```

plotly
```
px.scatter(df, x='price', y='units')
```

WATCH Dots on top of each other? Add `alpha=0.5` to see them.

Share of a whole? PIE, 2 TO 4 PARTS

Show one part's share of a total.

matplotlib
```
p = df.groupby('segment')['revenue'].sum()
ax.pie(p, labels=p.index, startangle=90)
```

plotly
```
px.pie(df, names='segment', values='revenue')
```

WATCH Only when the message is one part's share. Else use bars.

What is a total made of? STACKED BARS

Split each total into its parts.

matplotlib
```
w.plot.bar(stacked=True, rot=0, ax=ax)
```

plotly
```
px.bar(g, x='region', y='revenue',
    color='segment')
```

WATCH Only the bottom part is easy to compare. Put the key part there.

FIX-THIS-CHART CHECKLIST

THE MESSAGE

- [ ] **Can you say the message in one sentence?**
  If not, explore further before you polish.
- [ ] **Does the chart type fit the question?**
  Find the question on page 1.
- [ ] **Can a reader get the point in five seconds?**
  Make the message the title. Highlight one thing.

```
ax.set_title('Laptops earn the most')
```

- [ ] **Are the bars in a useful order?**
  Sort them. Long names? Use `barh()`.

```
ax.invert_yaxis()
```

CLUTTER

- [ ] **Would deleting a mark lose anything?**
  Remove frame lines, grids and ticks with no data.

```
ax.spines[['top', 'right']].set_visible(False)
```

- [ ] **Does the chart need its legend?**
  Write labels on the data instead.
- [ ] **Is the number axis still needed?**
  Write the values on the bars, then hide it.

```
ax.bar_label(ax.containers[0], fmt='${:,.0f}')
ax.xaxis.set_visible(False)
```

HONESTY

- [ ] **Does every bar chart start at zero?**
  Remove the axis limit. Columns: fix the bottom.

```
ax.set_ylim(bottom=0)
```

- [ ] **Is every axis labeled with its unit?**
  Say so for a log scale. Avoid a second y-axis.

```
ax.set(xlabel='', ylabel='Revenue ($)')
```

- [ ] **Do you know why the dates start and stop there?**
  Show the whole period. Shade the part you mean.
- [ ] **Is every number in the title true?**
  Compute it in a variable, then re-check it.

```
top = t['revenue'].max()
ax.set_title(f'Top product: ${top:,.0f}')
```

- [ ] **Is the chart flat?**
  No 3D, tilt or shading for effect.
- [ ] **Does the message survive other limits or dates?**
  Tell the reader which choice you made.
- [ ] **Is all the data shown?**
  Cut an axis or left rows out? Say so on the chart.

THE READER

- [ ] **Does it work in gray and for colorblind readers?**
  Vary lightness, line style or labels, not only hue.
- [ ] **Can a colleague say what it shows?**
  Their sentence should match yours. If not, fix the title.
- [ ] **Does the saved file look like the screen?**
  Save with `bbox_inches='tight'` (see Setup & save).

RULES OF THUMB

1. **Message first**
   Write it in one sentence and make it the title.
2. **Bars start at zero**
   A line may start higher if the title gives the size of the change.
3. **Sort bars by value**
   Unless the order is natural, such as months.
4. **Pies: 2 to 4 parts**
   And only when one part's share is the message.
5. **Four colors at most**
   More groups? Gray plus one highlight. Never 3D.
6. **Exact values? A table**
   Readers exploring on a screen? Use plotly.

MORE CHARTS

Total and parts over time?	**Stacked area**	`px.area()`
The trend in noisy data?	**Smoothed line**	`s.rolling(7).mean()`
What range did values cover?	**Shaded band**	`ax.fill_between()`
Every value of a small table?	**Strip plot**	`sns.stripplot()`
What shape is the spread?	**Density, violin**	`sns.violinplot()`
Which way does a trend run?	**Trend line**	`sns.regplot()`
Columns that move together?	**Pair plot**	`sns.pairplot()`
Where is a grid high or low?	**Heatmap**	`sns.heatmap()`
Same chart for each group?	**Small multiples**	`px.bar(facet_col=)`
Does it follow geography?	**Map**	`px.choropleth()`
A total made of groups?	**Treemap**	`px.treemap()`
Where do we lose people?	**Funnel**	`px.funnel()`
Who does what, and when?	**Timeline**	`px.timeline()`
How did we get from A to B?	**Waterfall**	`go.Waterfall()`

SETUP & SAVE

Clean look	`sns.set_theme(style='ticks')`
Chart size	`fig, ax = plt.subplots(figsize=(6, 3.5))`
Dollar ticks	`ax.yaxis.set_major_formatter('${x:,.0f}')`
Show it	`plt.show()`
Save a PNG	`fig.savefig('chart.png', dpi=200, bbox_inches='tight')`
seaborn chart	`ax.figure.savefig('chart.png', dpi=200, bbox_inches='tight')`
plotly look	`fig = px.line(m, x='month', y='revenue', template='simple_white')`
Save plotly	`fig.write_html('chart.html')`